



Caden Whalen

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OBJECTIVE

Motivated Aerospace and Electrical Engineering student with a strong skill set in communication and problem-solving, specializing in avionics. Hardworking and adept at managing complex projects and collaborating with others across technical teams. Founder of a specialized microsatellite student organization centered around providing and testing cost-effective, open-source design solutions. Seeking internship opportunities in the general avionics field starting May 2027.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Expected Graduation: May 2029

Bachelor of Science in Aerospace Engineering

GPA: 4.00/4.00

Bachelor of Science in Electrical Engineering

GPA: 4.00/4.00

Concentration in Circuit Technology and Robotics

EXPERIENCE

Ramblin' Rocket Club (HAB), Atlanta, GA

August 2025 – May 2026

Responsible Engineer

High Altitude Balloons at Georgia Tech is a project team of the Ramblin Rocket Club where members work towards the launch and recovery of a payload attached to a high-altitude weather balloon.

- Responsible for the design and development of a low-voltage nichrome wire cutdown device capable of generating 50 pounds of force, providing a reliable mechanism to prevent the payload entanglement during retrieval operations.
- Created and integrated a new payload bay tailored to accommodate custom-soldered electronics and GPS hardware with the goal of optimizing mounting interfaces and cable routing to ensure reliable system integration and payload operation.

Yellow Jacket Space Program (YJSP), Atlanta, GA

January 2026 – Present

Avionics Team Member

Student organization working to design, construct, and fly a liquid-fueled rocket to the Karan Line (100 km), the official edge of space.

- Responsible for addressing a faulty sensor; designing, creating, and integrating a sensor controller PCB with proper electrostatic discharge protection, reverse polarity protection, and proper filtering; resulting in the restoration of full sensor functionality.
- Positioned to become a responsible engineer by Fall 2026.

Georgia Tech Orbital Initiative (GTOI), Atlanta, GA

August 2026 – Present

Co-Founder, President

Student organization aiming to advance access to space technology through the design, development, and testing of cost-effective, open-source microsatellites.

- Identified a lack of support for undergraduate students interested in microsatellite technology and am currently founding a specialized student organization focused on developing accessible, cost-effective, and innovative satellite systems.
- Leading the development of three microsatellite technology projects focused on conventional satellite architectures, low-cost microsatellite platforms, and next-generation thermal-management solutions for sustaining AI computing in space.
- Positioning to become officially recognized by Georgia Tech in Spring 2027.

PROJECTS

Miniature Ant Robot for Search and Rescue, Atlanta, GA

August – December 2025

Frame & Battery Lead

Team-based project tasked with designing a miniature robot capable of locating and aiding in the rescue of victims trapped in collapsed buildings during natural disasters.

- Responsible for designing the carbon-fiber frame and retractable battery mechanism as well as integrating a servo-powered 360-degree camera, IoT audio network, and LED lights, ultimately resulting in over 40 unique, interlocking components.

SKILLS

Programming: C, Glowscript, Java, MATLAB, Python, RISC-V Assembly

Software: Altium Designer, GitHub (HTML), Linux (Kali/Ubuntu), Multisim, OpenRocket, Quartus Prime, SolidWorks

Hardware: Ammeters, Bread Boards, Oscilloscopes, PCB Prototyping, Strain Gauges, Voltmeters, Wind Tunnel Testing